Východiska a cíle AOL

BAM31AOL Aplikovaná elektronika v lékařství

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Biomedical Engineering: A Definition

- “Bioengineering is the broad umbrella term used to describe this entire field, bioengineering is usually defined as a basic research-oriented activity closely related to medicine, biotechnology and genetic engineering.”
- “It is clear that bioengineers of the future will have a tremendous impact on the quality of human life.”

Pacela: Bioengineering Education Directory. Quest Publishing Co., 1990

Biomedical Engineering: A Definition

- „Biomedical engineering (BME) or medical engineering is the application of engineering principles and design concepts to medicine and biology for healthcare applications (e.g., diagnostic or therapeutic purposes). BME is also traditionally logical sciences to advance health care treatment, including diagnosis, monitoring, and therapy. “

Wikipedia: Biomedical Engineering. URL: https://en.wikipedia.org/wiki/Biomedical_engineering. Feb 2024

Biomedical Engineering: A Definition

- „Biomedical engineering is a multidisciplinary field that combines engineering principles with biological sciences to develop technologies and devices for the diagnosis, treatment, and monitoring of health and diseases. The goal is to improve patient care and understand biological processes at the level of systems and tissues.“

ChatGPT 3.5, Feb 2024

- answer of the ChatGPT 3.5 machine on the input „Can You define, please, in one or two sentences, what is it biomedical engineering?“

Applied Optoelectronics in Medicine

- and one more question „What do You imagine under the term Applied optoelectronics in medicine?“
- „Applied optoelectronics in medicine involves the use of optical and electronic technologies for the diagnosis, treatment, and monitoring of patients. This can include optical imaging, laser systems, optical sensors, and other technologies for tissue imaging, surgical procedures, non-invasive monitoring, and other medical applications.“

ChatGPT 3.5, Feb 2024

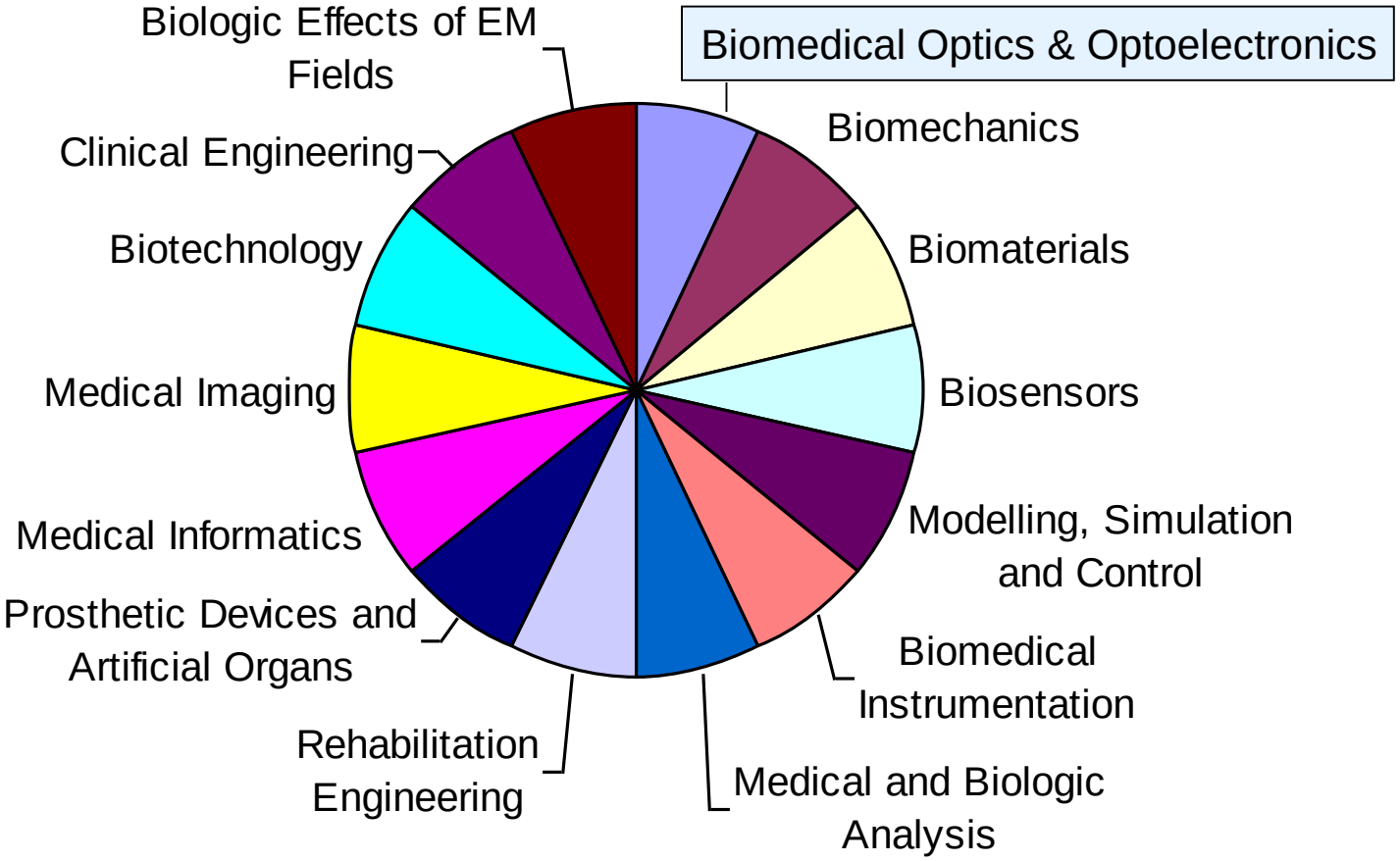
Applied Optoelectronics in Medicine

- the lectures are prepared with the kind support of

Prof. Dr.-Ing. Vladimír Blažek, Dr.h.c.
Chair for Medical Information Technology (MedIT)
Helmholtz-Institute for Biomedical Engineering
RWTH Aachen University

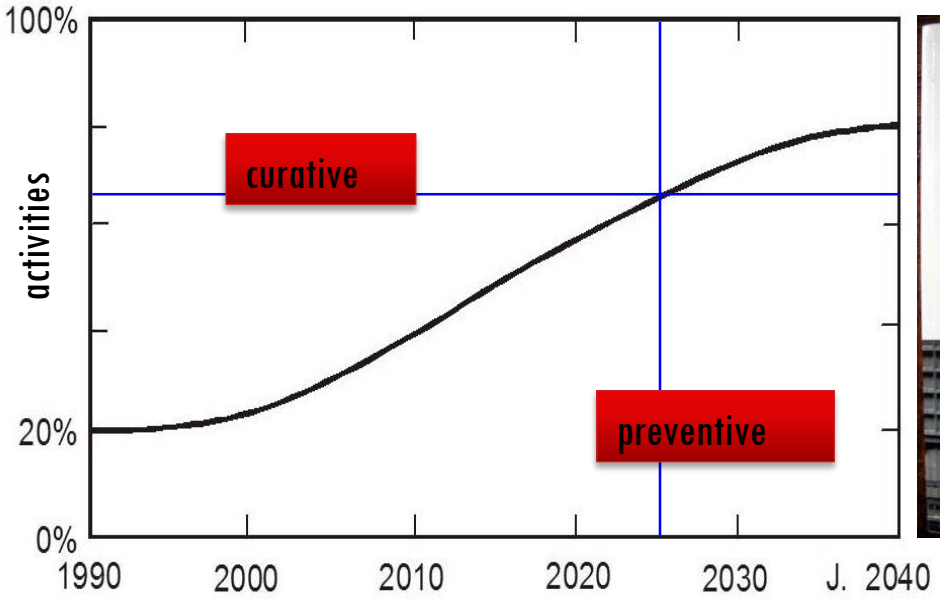
2002 Honorary Doctorate of FEL CTU
2016 Emeritus Professor of RWTH Aachen
member of the Scientific Council of FEL CTU and CTU

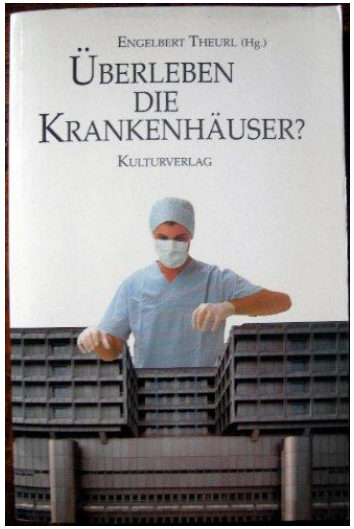
The Disciplines of Biomedical Engineering

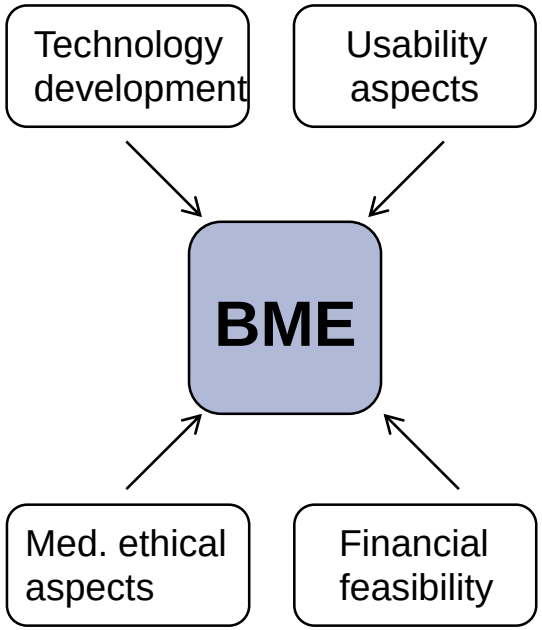


Current trends in Medicine „3N“-Diagnostics

- non-invasive,
- non-burden,
- non-expensive acquisition of functional biosignals



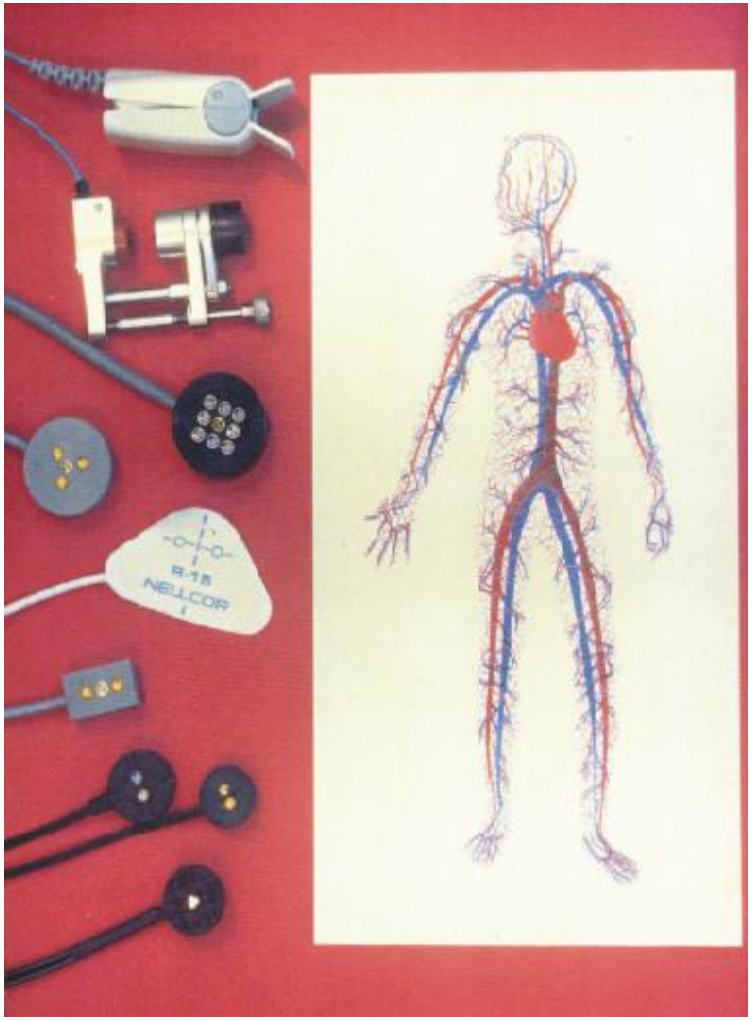




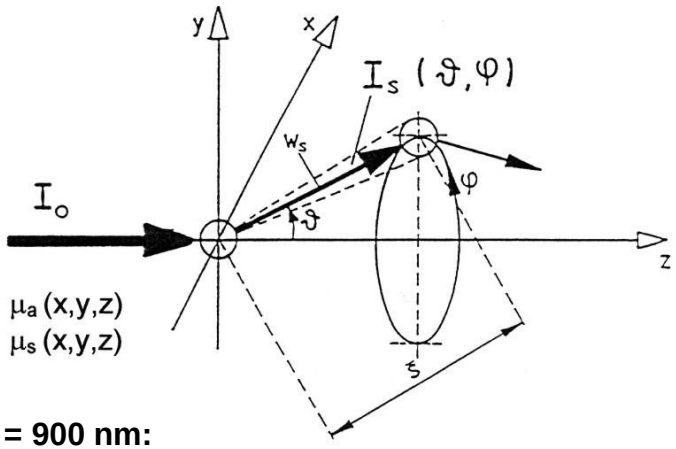
E. Theurl (10991): Überleben die Krankenhäuser?

Fundamental questions on photon migration in bio-tissue

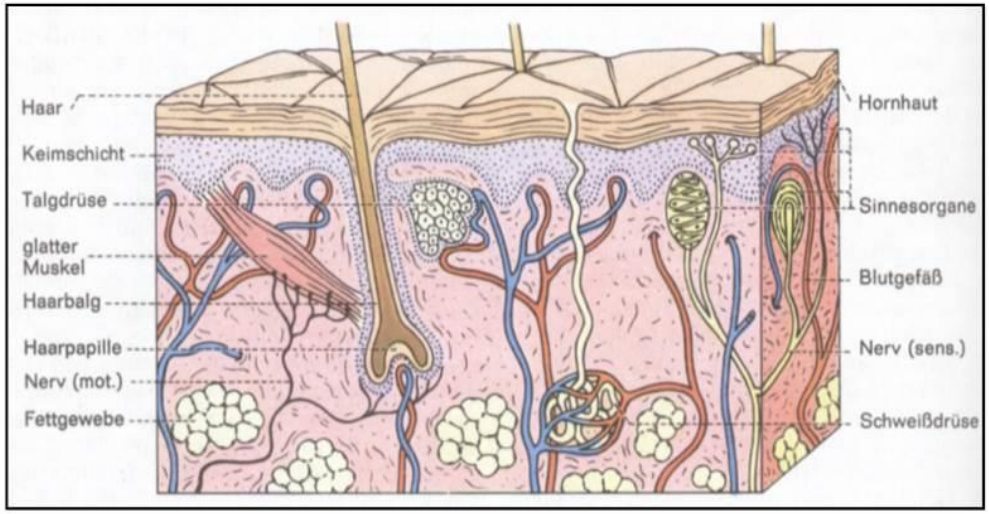
- How does optical radiation pass through high scattered biological objects?
- Which spectral range is useful?
- How does the detected radiation depend on the blood volume in transilluminated area?

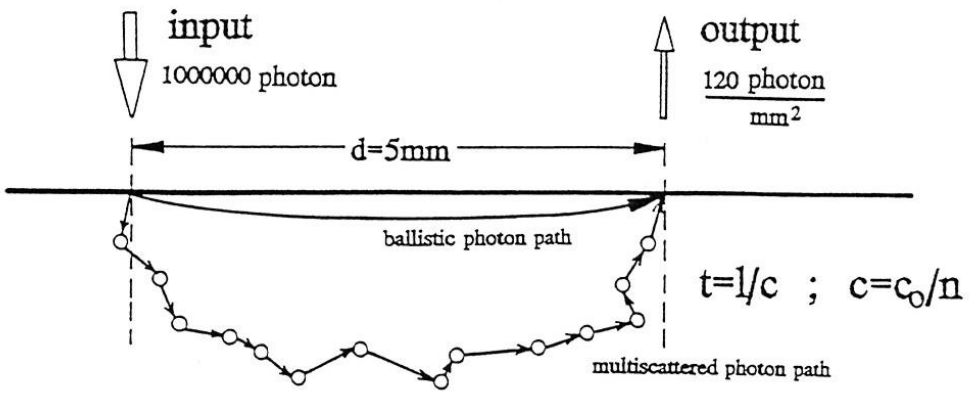


Light propagation in bio-tissue



λ = 900 nm:
mean free photon path: s = 0.24 mm
absorption probability : w_a = 2 % per collision
scattering probability : w_s = 98 % per collision





Speed of light in tissue (by n=1.5)**:** **c = 2x10^8 m/s**
Shortest flight time of balllistic photons
(by l=d=5 mm)**: t = 25 ps**

Remember:

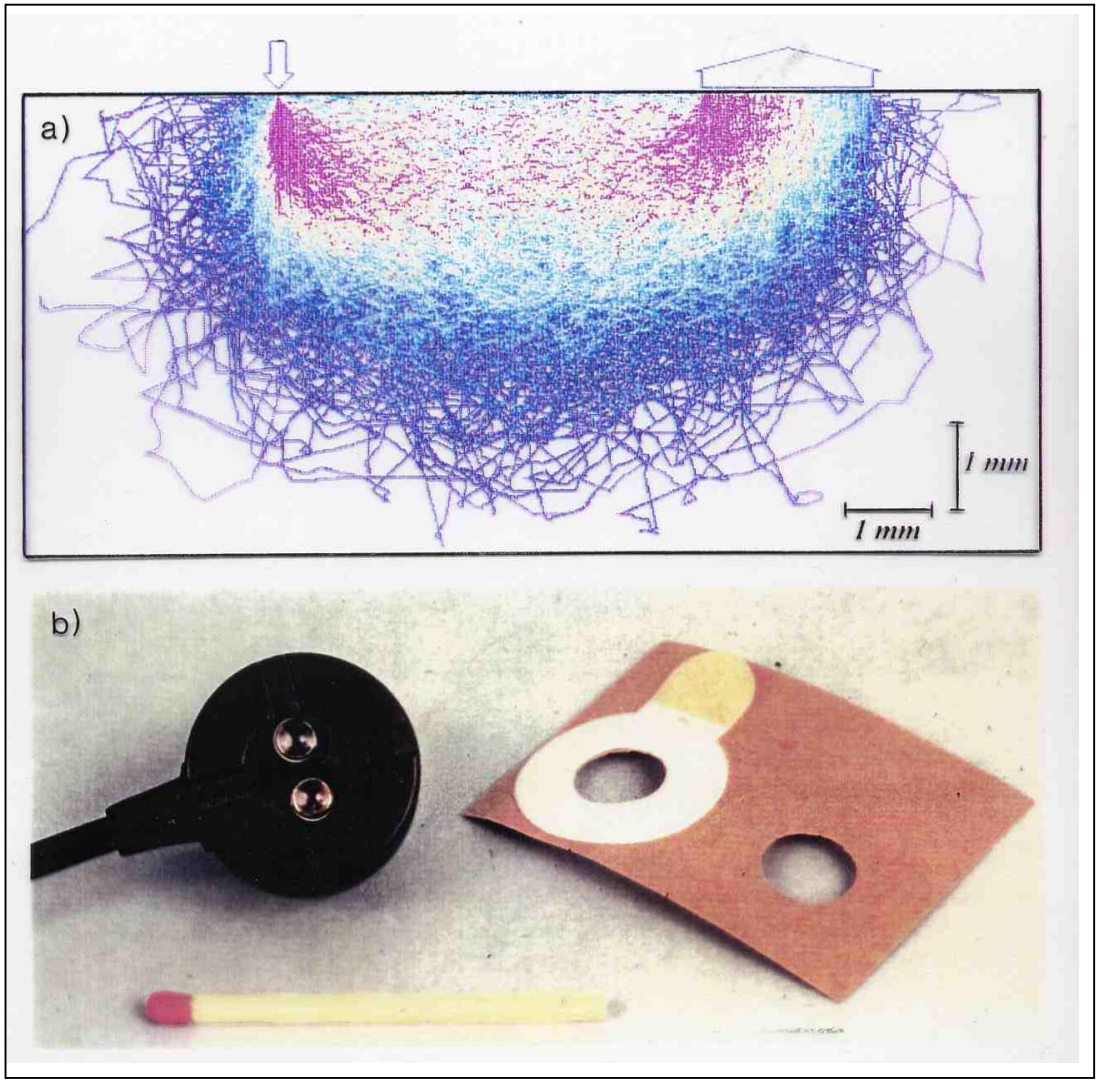
Probability of a photon scattering in bio-tissue is about 50 times higher as the probability of absorption.

“Ray tracing” simulation (Monte-Carlo Method) in optical range

Infrared light distribution in skin-phantom simulated for a classical optoelektronic sensor (Fig. **b**, 1x IR-LED; 1x Si-PD; both with lenses)

Distance Light source-Photodetector: 5 mm
Penetration depth: up to aprox. 5mm at λ = 900 nm
Volume of P/T interaction: aprox. 150 mm^3

Shadow-projection of scattering photons which achieve the active sensor area (1516 from $3 \cdot 10^6$ photons).

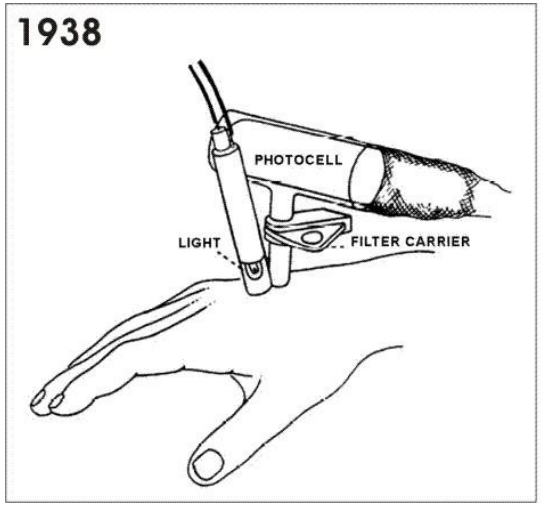


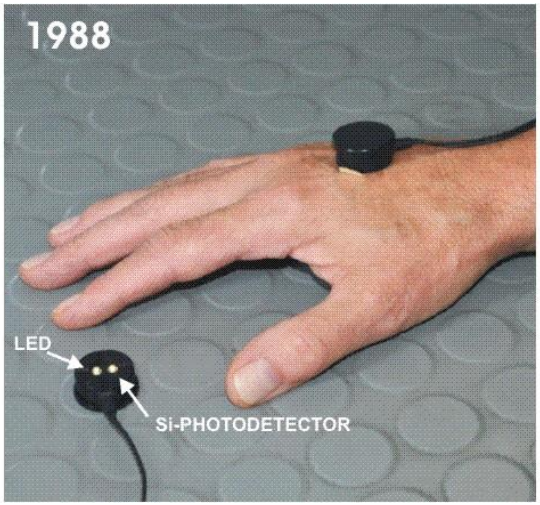
Digression: Skin effect

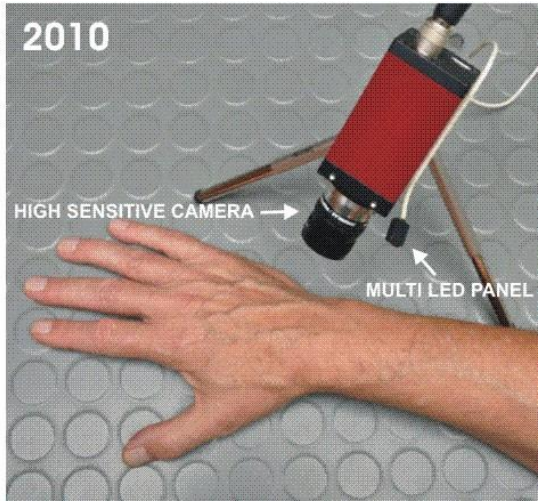
$$J = J_S \, e^{-d/\delta}$$

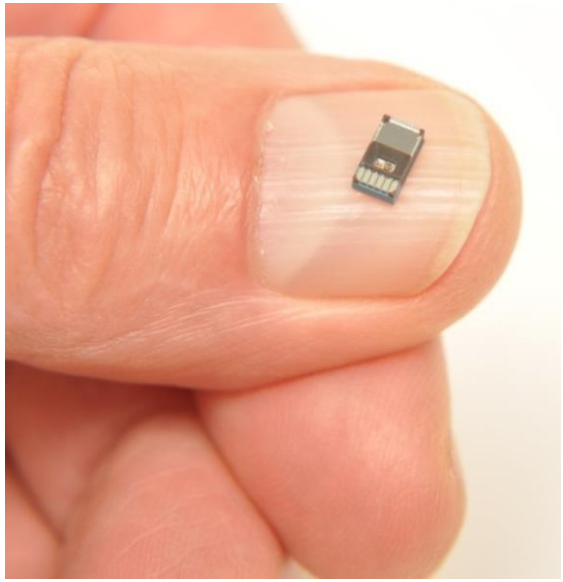
$$\delta = \sqrt{\frac{2\rho}{\omega\mu}} \sqrt{\sqrt{1 + (\rho\omega\varepsilon)^2} + \rho\omega\varepsilon}$$

Design of optoelectronic sensors and their characterization









Benefits of optoelectronic sensors in medical diagnostics

- non-invasive and not disturbing method,
- possibly automated, self-calibrating,
- free region of interest ,
- small, light, hand-manageable, easy integration of different sensors,
- low budget, disposable components,
- flexible and reproducible application, long-term measurements possible,
- multiple wavelengths recording possible (Optical Biopsy),
- contact-less signal acquisition,
- spatially resolved measurement possible (2D, Photoplethysmography Imaging)
- depth-resolved tissue-structure analysis possible (2D/3D,Optical Tomography).

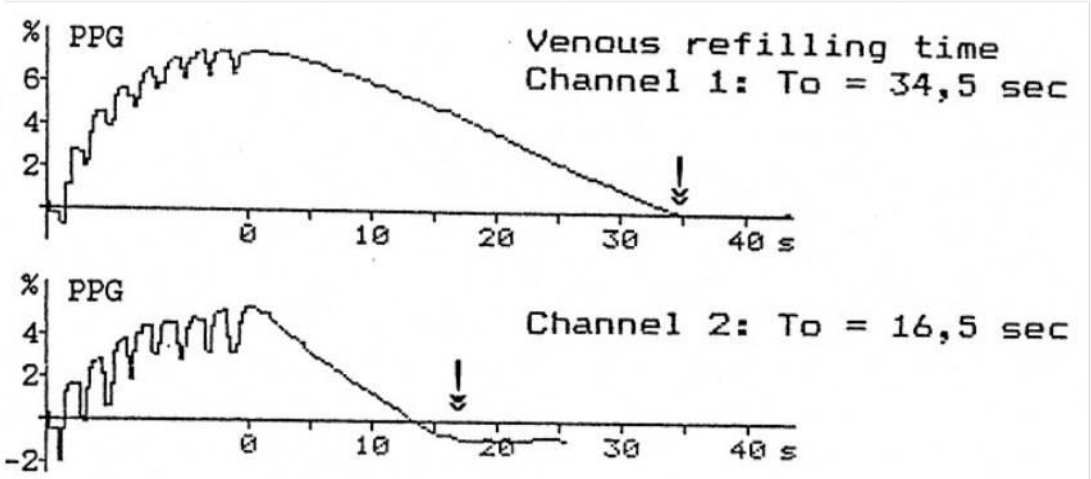
Remember:

Sensors are the hub and bridge of many BMI activities, they are germ cells for innovative system concepts in evidence based medical diagnostics

Some constructive implementations of optoelectronic sensors

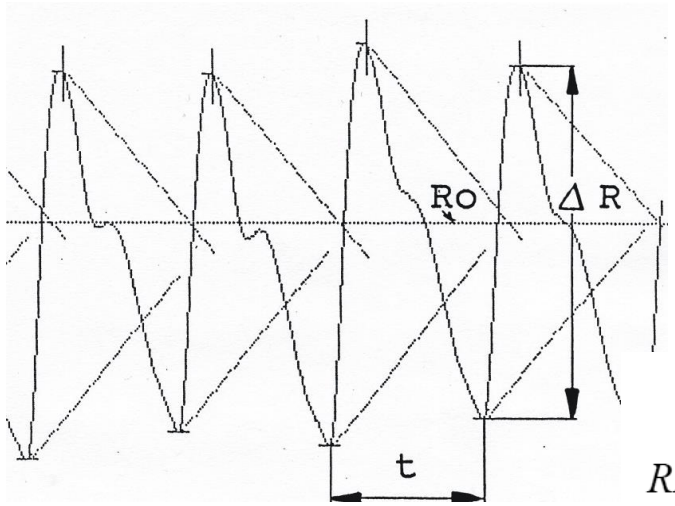
Photoplethysmography (PPG)

- Venous hemodynamics
  (PPG muscle pump test)



Photoplethysmography (PPG)

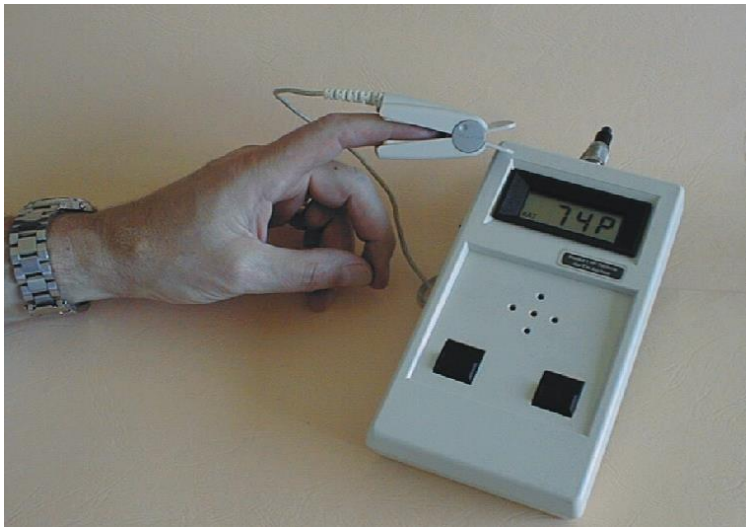
- Arterial hemodynamics (peripheral heart beat synchronous blood volume pulsation)



$$GPI = \frac{\Delta R}{t \cdot R_0}$$

$$\overline{RR} = \frac{\sum_{i=1}^{n} RR_i}{n}$$

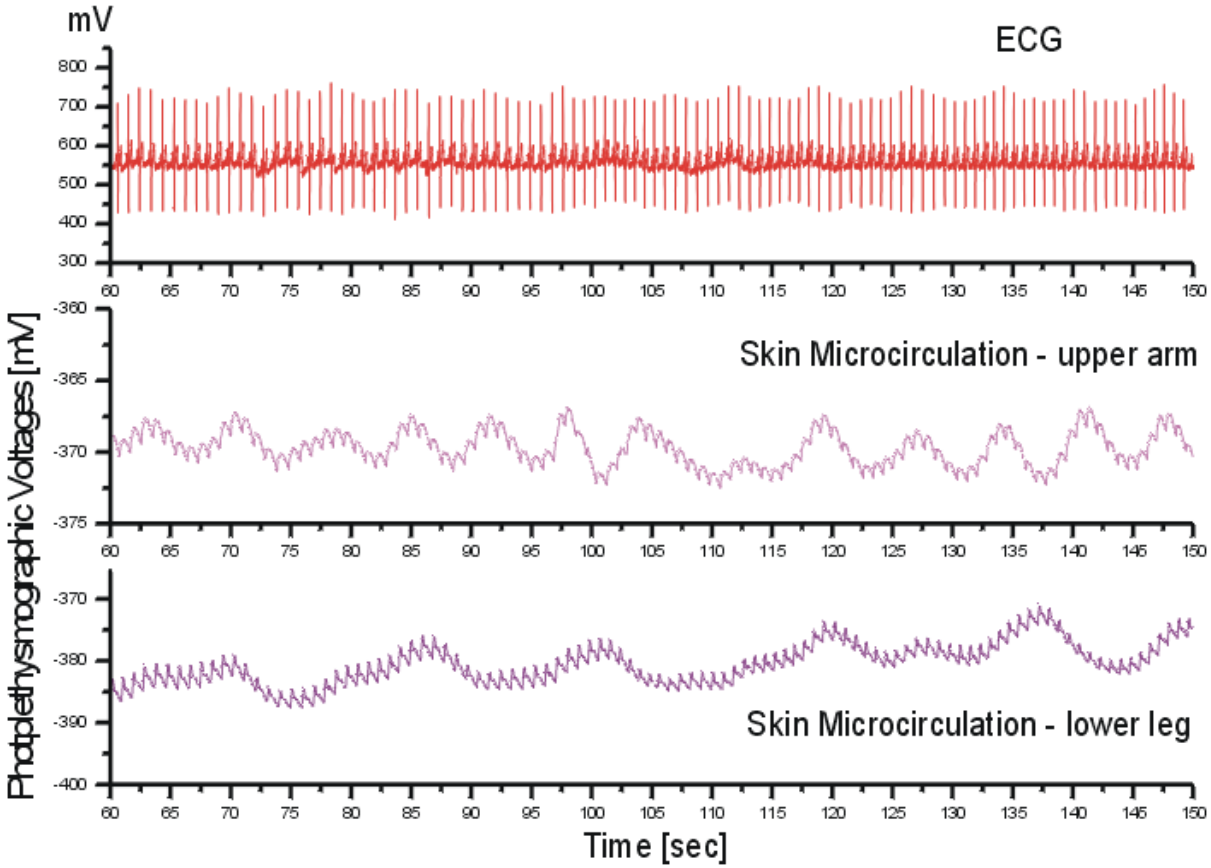
$$RR_{SD} = \sqrt{\frac{\sum_{i=1}^{n} \left(RR_i - \overline{RR}\right)^2}{n}}$$

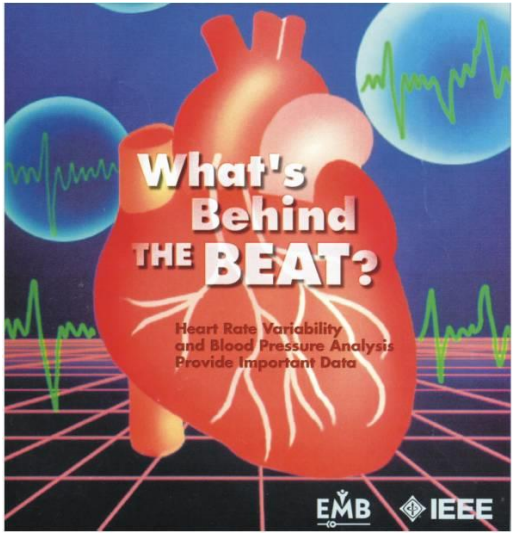


Remember:

- *During each heartbeat, blood is pushed by the vessels of the skin, and the extension of vessels under the OES determines the amount of light reflected / scattered back.*
- *Peripheral arterial blood volume pulse assessment is well used in medical diagnostics, especially for the heart rate (HR) and heart rate variability (HRV) analysis by cardiovascular risk patients, but for example also as a bio-feedback control signal in stress and pain diagnostics.*

PPG assessment of dermal perfusion rhythmicity





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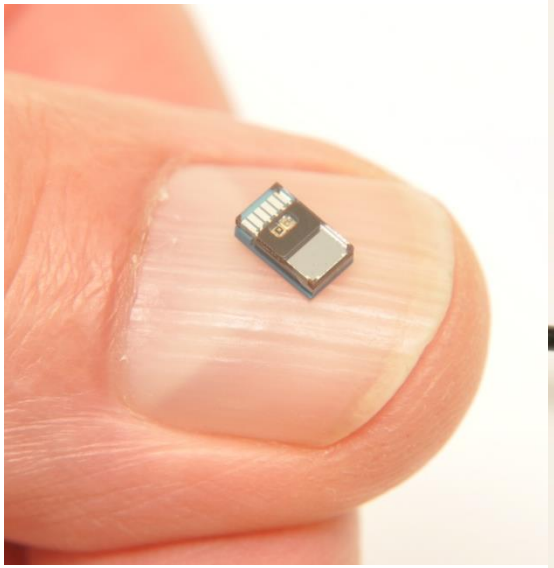
Vol. 20, No 2 (2001)

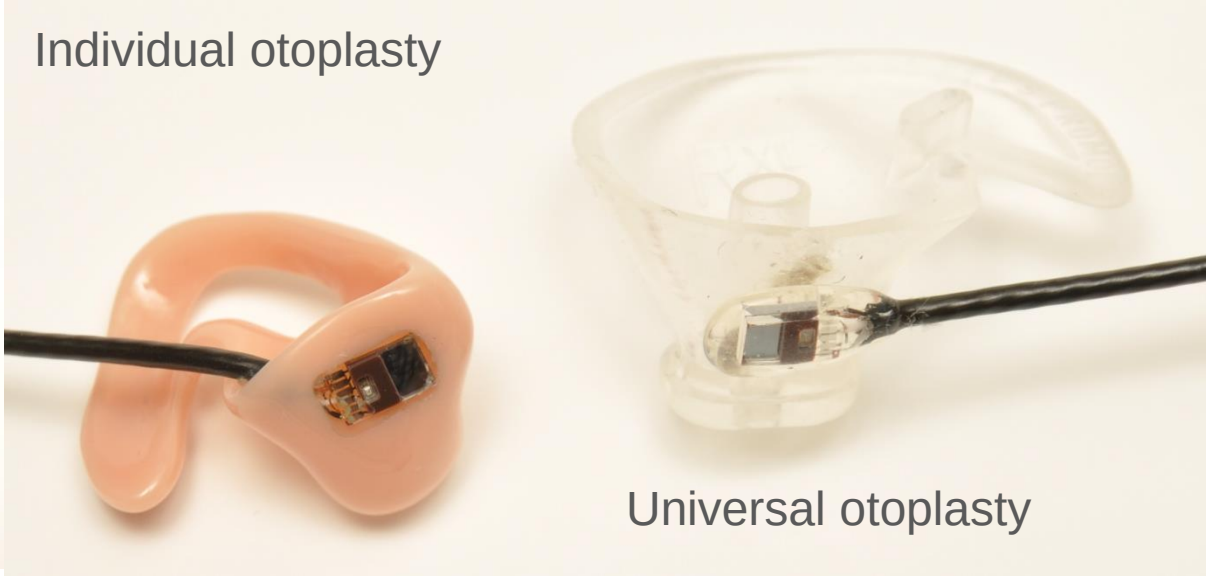
Remember:

Besides the HB and respiratory synchronous rhythms also other perfusion rhythms can be detected optoelectronically, their origin and their behavior is currently being investigated

In-ear OES measurement scenario

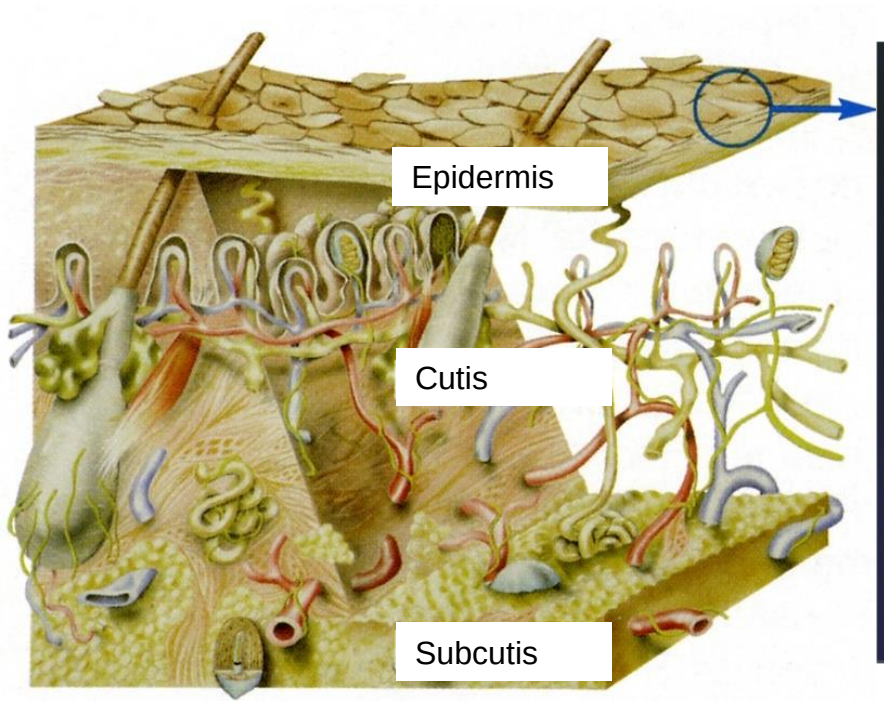
- Physiologically meaningful
- Metrologically advantageous
- Optimal application port
- Vital signs monitoring with micro optical sensor PX 01-2 (CiS, Erfurt)
- Multi -λ-PPG-sensor in silicon-glass-ceramic technology
- Spectral sensitivity at 850 nm: 0.55 A/W
- Optical crosstalk at 640 nm: $<10^{-3}$
- Sensor dimensions: 8 x 4 x 1.5 mm^3

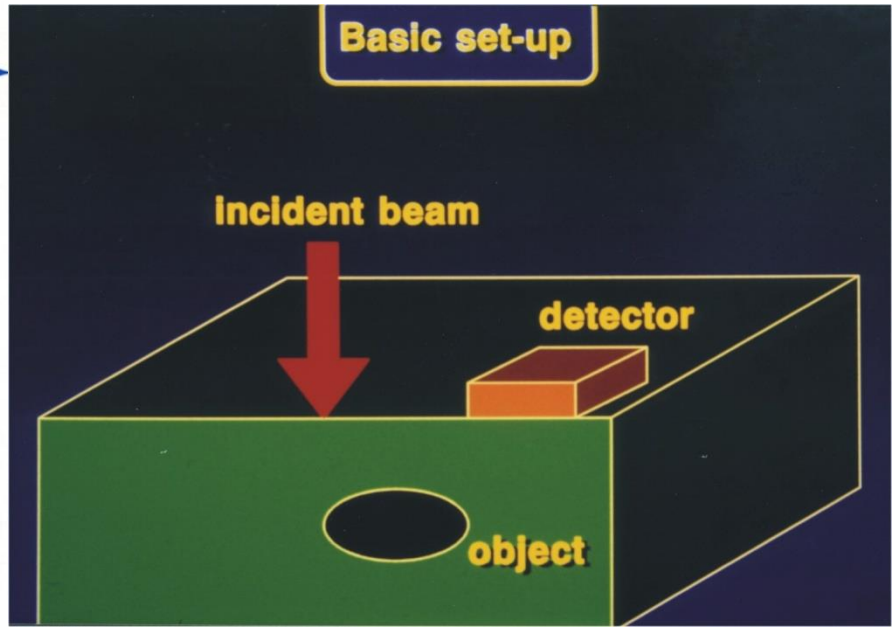




Optoelectronics imaging methods in medicine – „virtual images for virtuoso diagnostics“

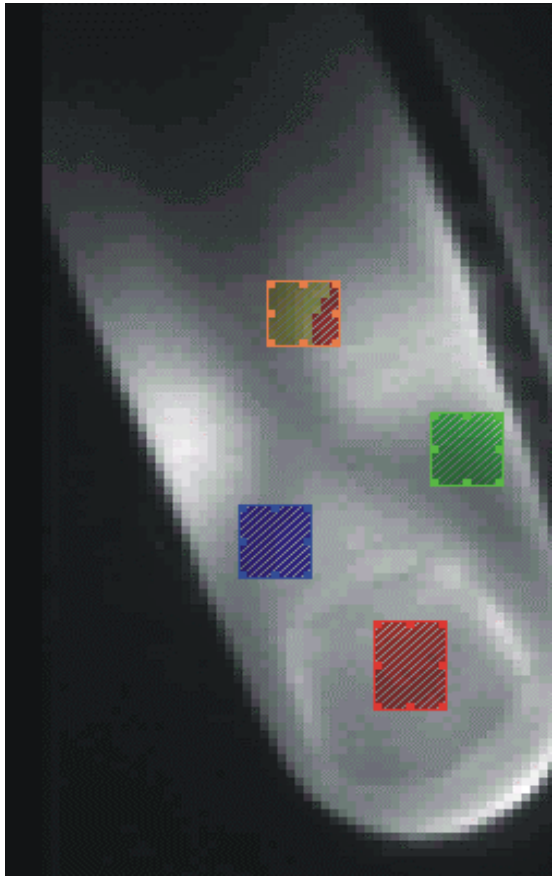
- Photoplethysmography Imaging (PPGI)
- Infrared Thermography (IRT)
- Optical Coherence Tomography (OCT)

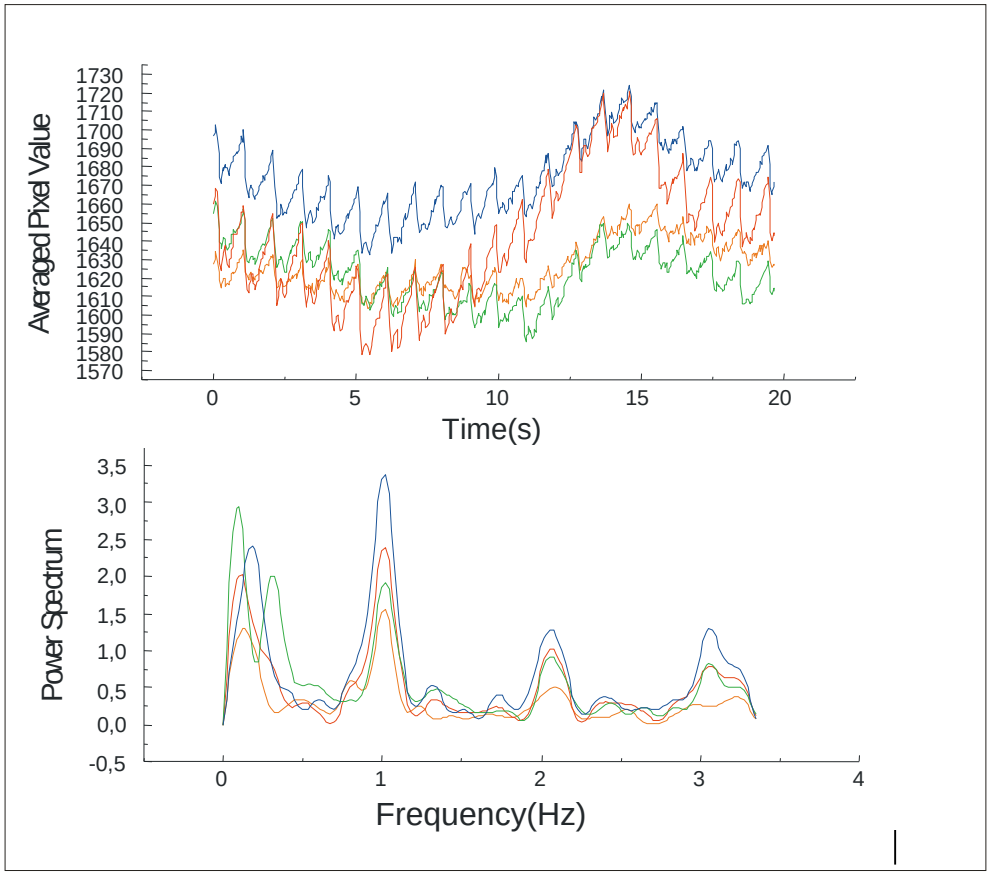




Photoplethysmography Imaging (PPGI)

- Example 1: Monitoring of distributed blood perfusion status in healthy skin



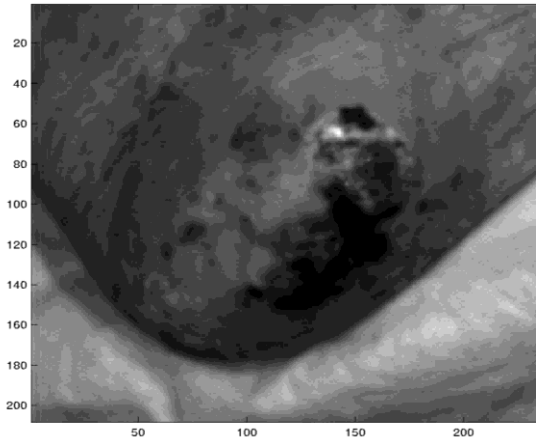


Photoplethysmography Imaging (PPGI)

- Example 2: Skin tumour perfusion status

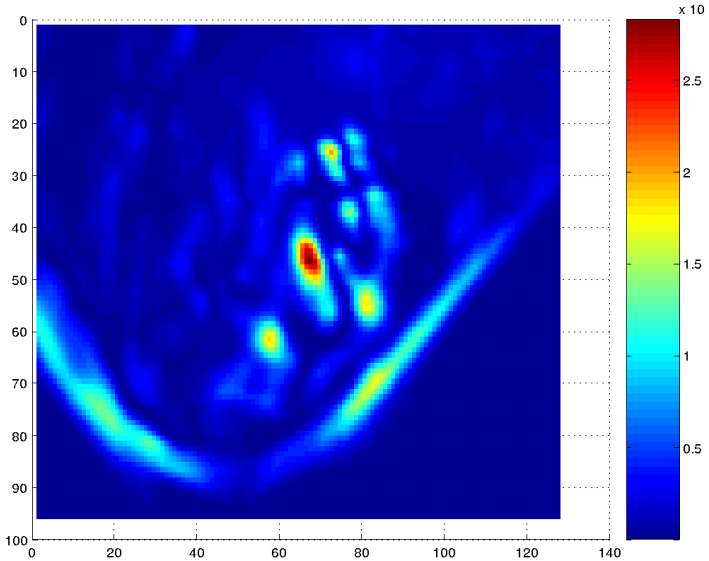
Standard CCD camera

PPGI image capture from a picture sequence of 900 shots



spatial resolution 50×50 μm, sampling rate15 fps, bit-depth14 bit

Virtual perfusogramm; the mean amplitude of the arterial (heart-synchronous) blood volume pulse is color-coded



Photoplethysmography Imaging (PPGI)

- Example 3: Latest perfusion studies using PPGI (RWTH Aachen)

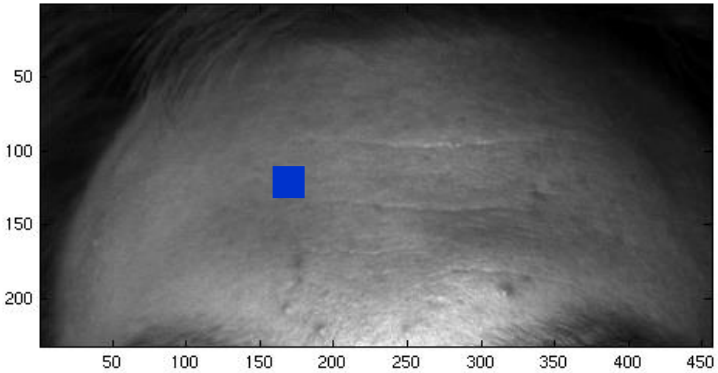
Standard image

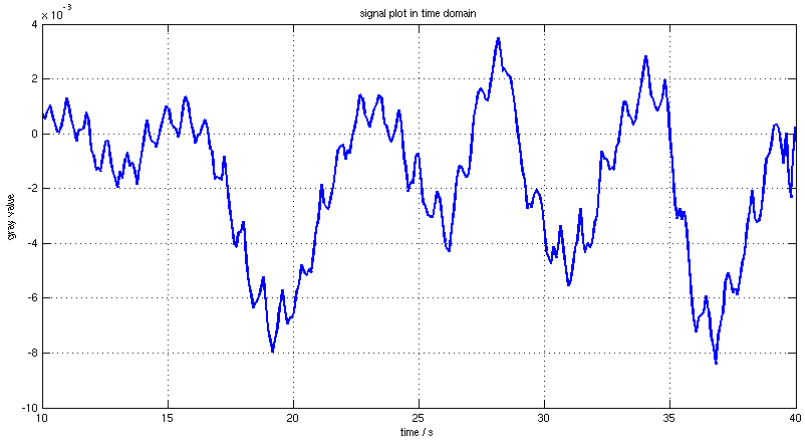
Post processed PPGI video

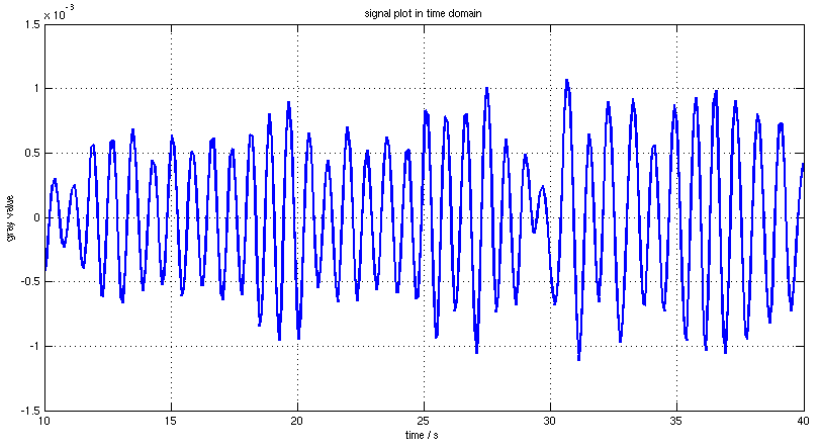
Photoplethysmography Imaging (PPGI)

- Example 3: Latest perfusion studies using PPGI (RWTH Aachen)

one ROI (virtual PPG sensor),
16 x 16 pixel = 5 x 5 mm



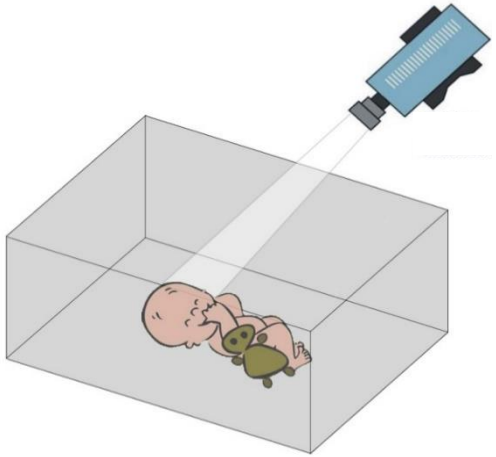


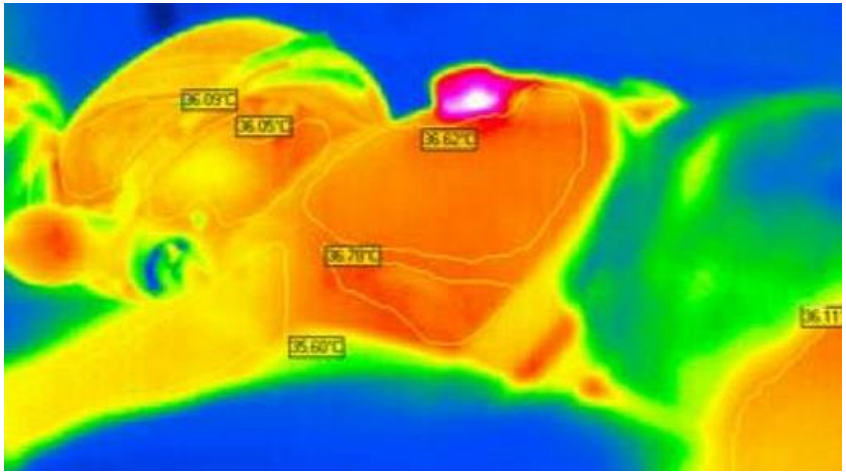


Infrared Thermography (FIRT)

- sensing modalities in today's incubator

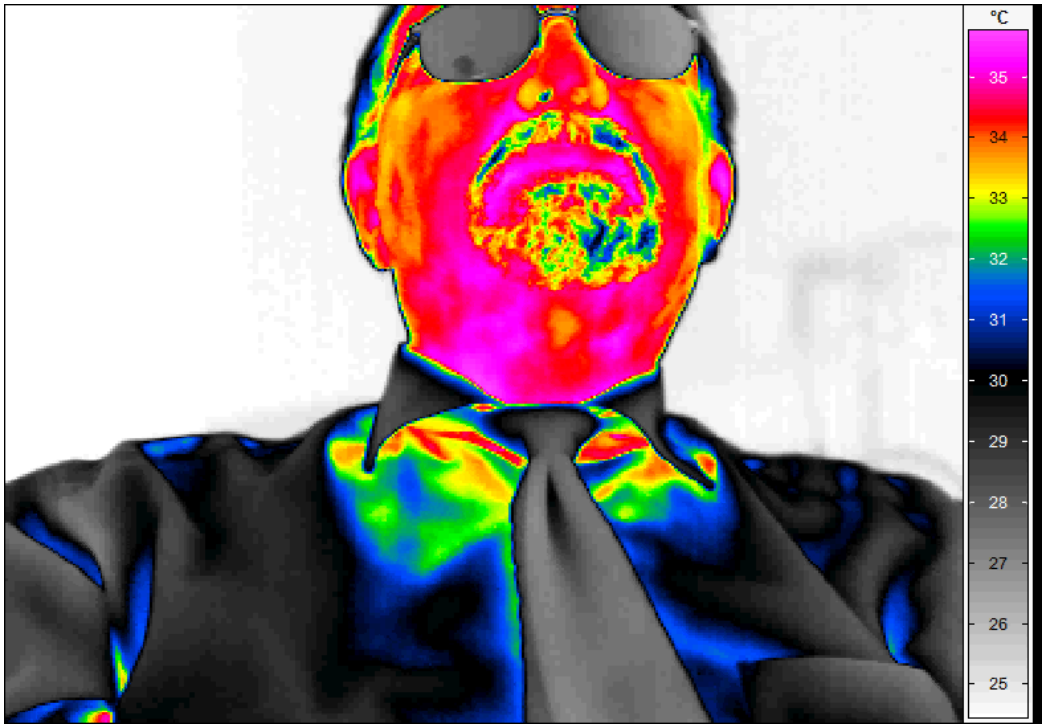
- a vision: smart incubator with wireless sensing





FIRT non-contact breathing detection

Model Opti-Res Clinic
(Jenoptik, Germany)
www.inframedic.de



AlGaAS/GaAs **QWIP**
(Quantum well infrared photodetector)

Spatial resolution: 384 x 288 pixel

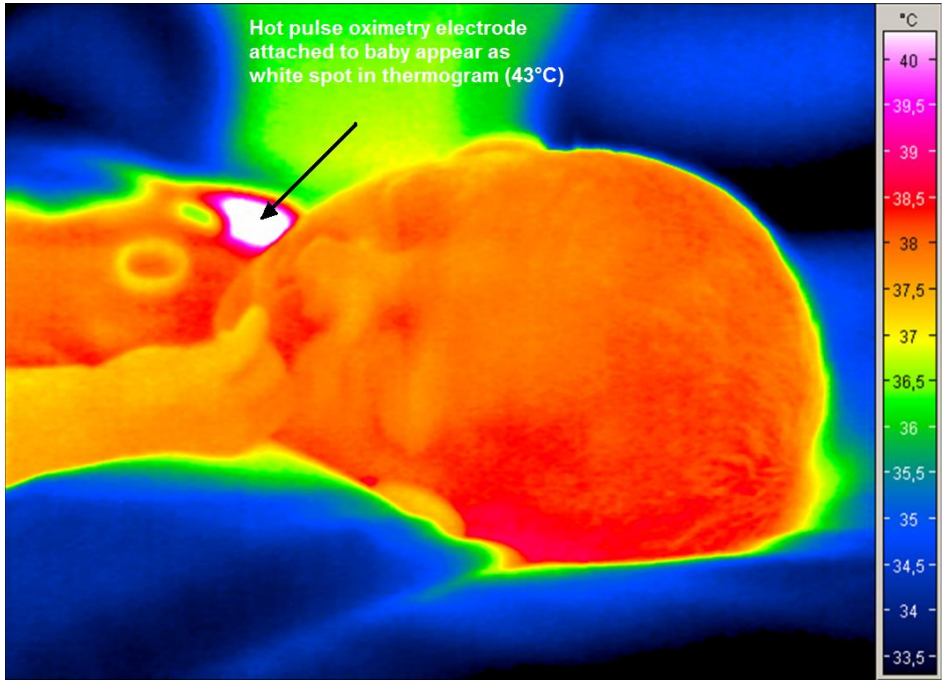
Thermal resolution: 0.05 °C

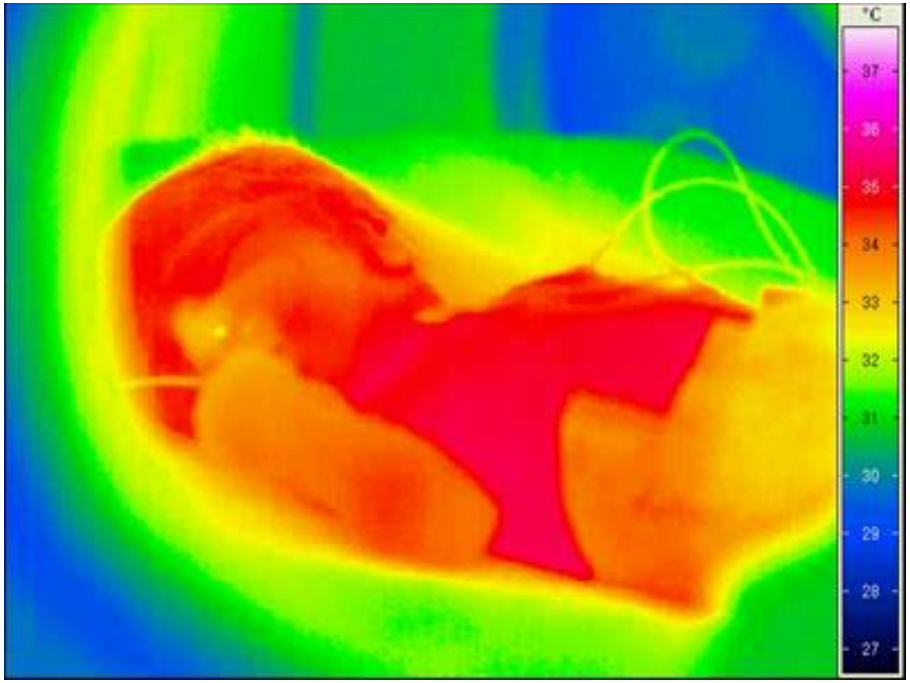
Time resolution: 10 fps

Thermal range: -40 °C – 1.200 °C

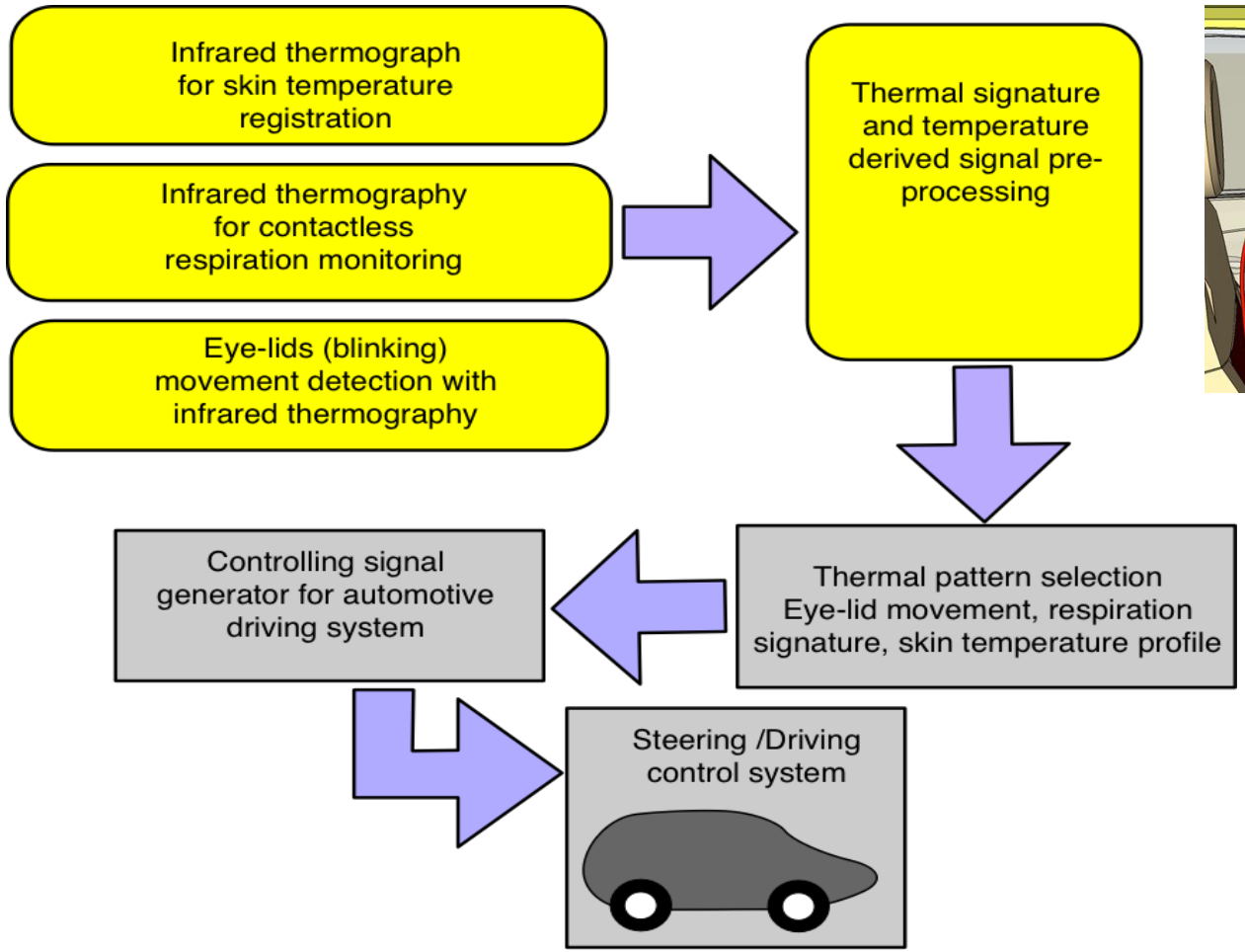
Example of neonatal breathing detection

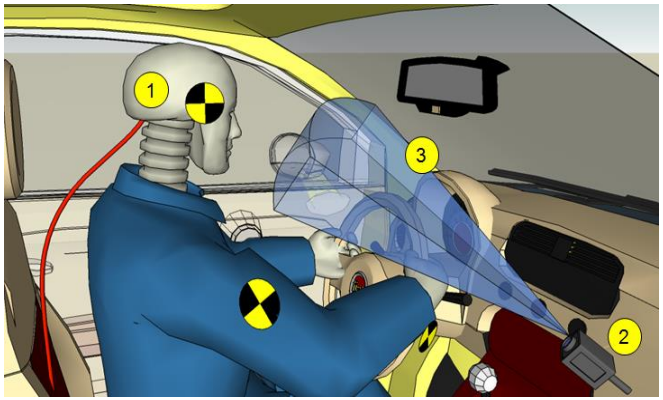
Example video

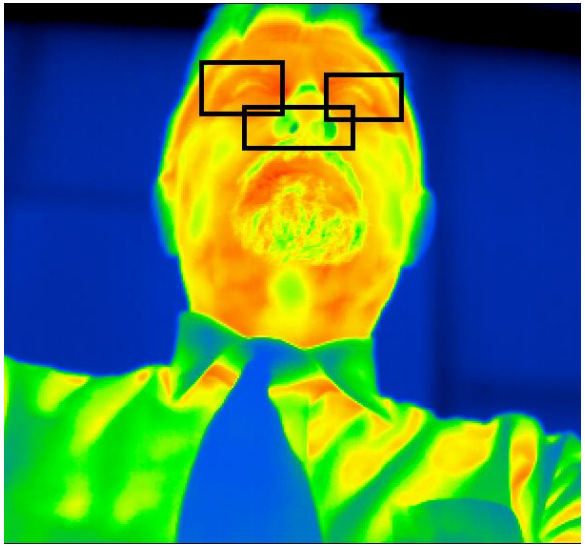




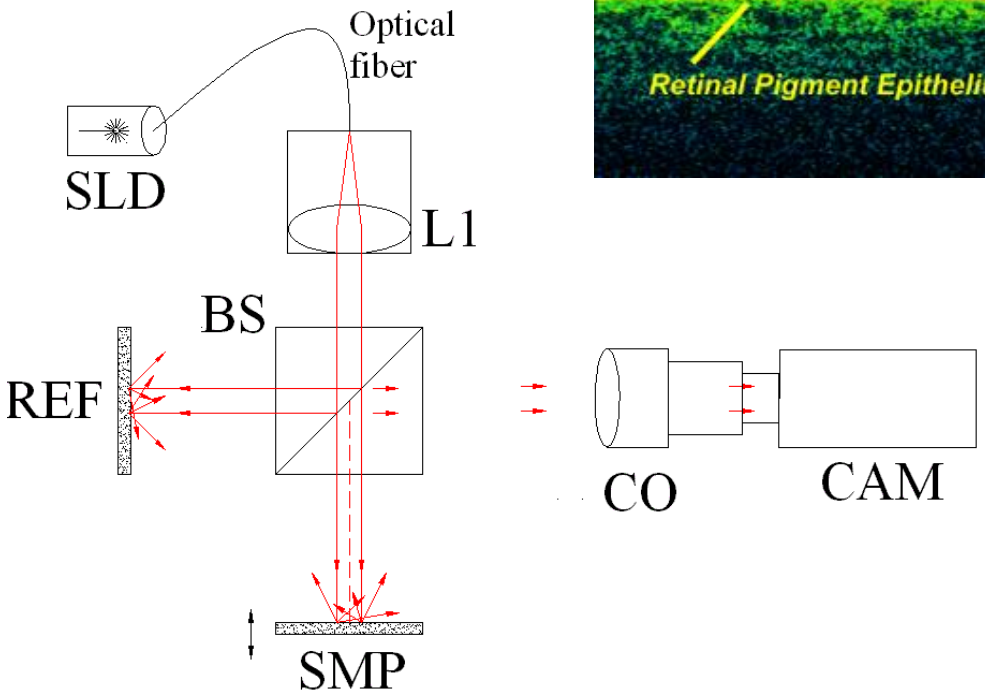
Automotive applications of FIRT

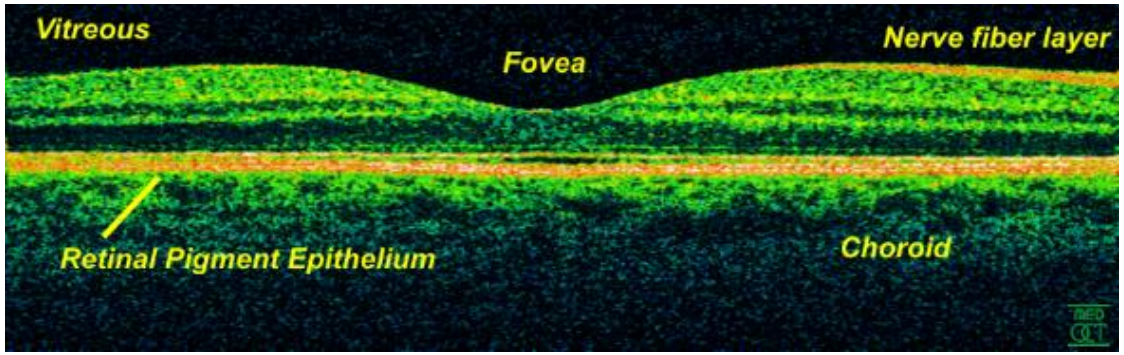






Optical Coherence Tomography (OCT)





Optical Coherence Tomography (OCT)

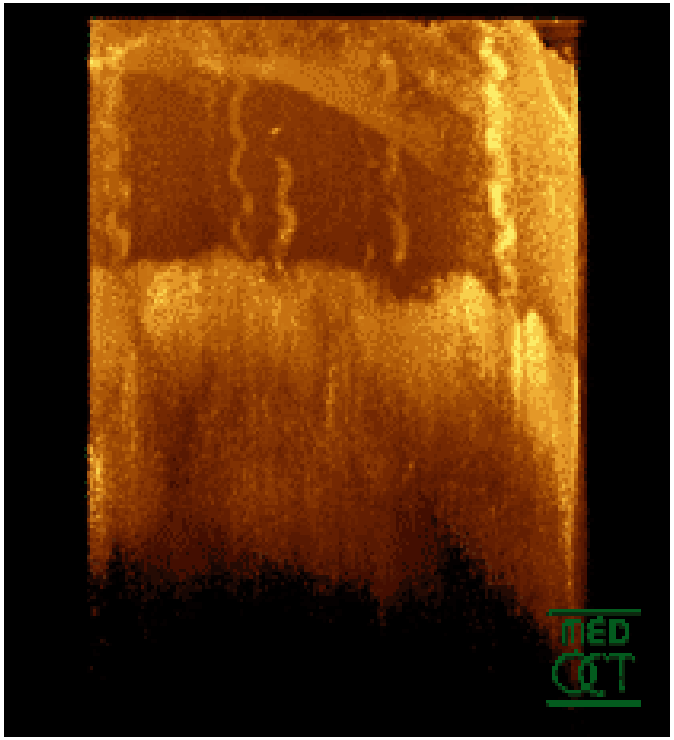
- optical coherence tomogram of the fingertip with sweat transitions in the dermis

Remember:

Optical coherence tomography (OCT) *is an optical signal acquisition and processing method. It captures micrometer-resolution, three-dimensional images from within optical scattering media (e.g., biological tissue).*

OCT is an interferometric technique, typically employing near-infrared light. The use of relatively long wavelength light allows it to penetrate into the scattering medium. Confocal microscopy, another similar technique, typically penetrates less deeply into the sample.

Depending on the properties of the light source (super LEDs, ultrashort pulsed lasers and supercontinuum lasers have been employed), OCT has achieved sub-micrometer resolution (with very wide-spectrum sources emitting over a ~100 nm wavelength range).



Side length 1x1 mm
Depth cca. 600 μm
Axial resolution cca. 5μm
Source: https://en.wikipedia.org/wiki/
/Optical_coherence_tomography